

Kyle Heller

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SUMMARY

Platform engineer with 8+ years in enterprise cloud environments, currently operating production Kubernetes on AWS EKS at a Fortune 75 financial services company. Background spanning platform operations, infrastructure automation, and technical leadership across multi-tenant cloud environments. Built internal tooling, automation pipelines, and self-service workflows using Python, API integrations, and workflow orchestration. Hands-on with Kubernetes cluster operations, Helm, ArgoCD, Cilium, observability (Prometheus/Grafana), policy enforcement (OPA/Gatekeeper), and container security in regulated environments. Certified Kubernetes Administrator, AWS Solutions Architect, Azure Solutions Architect Expert, and HashiCorp Terraform Associate. M.S. Computer Science from CU Boulder.

TECHNICAL SKILLS

Cloud & Infrastructure: AWS (EKS, EC2, IAM, S3, VPC, CloudWatch, RDS), Azure (AKS, Virtual Networks, Storage, Entra ID, Key Vault), Kubernetes, Docker, Terraform, Linux (RHEL, Ubuntu)

Platform & Orchestration: Helm, Kustomize, ArgoCD, Cilium (CNI/network policy), Gateway API, Pod Security Standards, OPA/Gatekeeper, Veeam Kasten, namespace isolation, resource quotas, cluster lifecycle management

CI/CD & Automation: GitHub Actions, Azure DevOps, ArgoCD (GitOps), Flux, n8n, Trivy, container image lifecycle

Development: Python, Go (reading proficiency), Bash, FastAPI, REST APIs, Rego (OPA), LLM API integration

Observability: Prometheus, Grafana, OpenTelemetry, ELK/EFK, Azure Monitor, CloudWatch, Log Analytics, SLO/SLI frameworks, alerting design

Security & Governance: IAM/RBAC, OPA/Gatekeeper (custom Rego policies), Pod Security Admission, Falco, secrets management, default-deny network policies, least-privilege design, container image scanning, PCI-DSS awareness

EXPERIENCE

Kubernetes Platform Engineer

May 2026 – Present

Nationwide — Columbus, OH

- Platform engineer on the Application and Database Platforms team, operating production EKS clusters on AWS supporting enterprise financial services workloads in a regulated environment.
- Kubernetes cluster operations including upgrades, node group management, Cilium CNI networking, pod security enforcement, and Gatekeeper policy development for compliance-driven workloads.
- Developed internal platform tooling and automation using Python and Bash to streamline operational workflows, improve deployment reliability, and reduce manual toil across the platform engineering lifecycle.

Technical Lead

Jul 2022 – Apr 2026

Veeam Software — Columbus, OH

- Led an 8-person engineering team supporting enterprise cloud infrastructure across multi-tenant environments; served as senior technical escalation point for complex issues spanning compute, storage, networking, and application layers.
- Built end-to-end automation for compliance reporting using n8n, Python, Puppeteer, and Claude API integration, reducing a 2–3 hour manual workflow to ~15 minutes and eliminating data accuracy issues.
- Conducted technical assessments of partner environments, identifying configuration risks and recommending architectural improvements tied to security, business continuity, and cost optimization.
- Served as cross-functional liaison between engineering, R&D, and QA teams, delivering structured root cause analysis with reproducible evidence to accelerate product fixes and improve system reliability.
- Mentored and coached a team of 10 engineers on troubleshooting methodology, technical analysis, and knowledge management; established standardized operational playbooks and diagnostic workflows as Knowledge Domain Expert.

Cloud Infrastructure Engineer

Jan 2019 – Jul 2022

Veeam Software — Columbus, OH

- Managed production cloud infrastructure for enterprise partners running BaaS and DRaaS environments; diagnosed and resolved failures across backup, replication, storage, and network layers to minimize downtime and data loss risk.

- Led solution design, migration planning, and infrastructure upgrades for enterprise partners; provided hands-on implementation for cloud platform configuration, multi-tenant onboarding, and environment deployments.
- Performed structured infrastructure analysis and environment reviews; translated complex technical findings into actionable recommendations for engineering and management stakeholders.

PROJECTS

Platform Starter Kit • Terraform + Kubernetes + Veeam Kasten + ArgoCD • github.com/kyle-heller/Platform-Starter-Kit

- Terraform modules provisioning AKS clusters with configurable node pools, networking, RBAC, and remote state on Azure Storage; separate dev/prod environment configurations. Kasten storage module provisions Azure Blob backend with managed identity RBAC for K10 exports.
- Veeam Kasten integration: K10 backup policy with daily snapshots, 7/4/3 GFS retention, and automated export to Azure Blob via managed identity; VolumeSnapshotClass with CSI annotation; OPA/Gatekeeper policy requiring StatefulSets to carry a k10.kasten.io/backup label, enforcing backup governance at the admission layer.
- ArgoCD GitOps delivery with automated sync, self-heal for drift detection, and prune for removed resources; Application manifests managing Kasten infrastructure and application deployments from Git as source of truth.
- Pod Security Standards (restricted profile), default-deny NetworkPolicies, Gateway API routing, Kustomize-based team onboarding with namespace isolation, resource quotas, RBAC bindings, and LimitRanges. Architecture decisions documented via ADRs.

UpDog Monitor • FastAPI + Prometheus + Grafana + React • github.com/kyle-heller/UpDog-Monitor

- Full-stack uptime monitoring service: FastAPI backend with PostgreSQL (Alembic migrations), React/TypeScript frontend, and background worker performing health checks. SLO engine computing availability (99.5% target) and latency (p95 < 500ms) with error budget burn rate tracking over a rolling 30-day window.
- Prometheus instrumentation with custom histograms, counters, and gauges; Grafana dashboards provisioned via JSON. Alerting rules for SLO breaches, error budget burn rate, and endpoint downtime using histogram_quantile and rate queries.
- CI/CD via GitHub Actions: lint, test, build, push to GHCR, and Trivy container image scanning on CRITICAL/HIGH vulnerabilities. Kubernetes manifests with Kustomization for cluster deployment.

OPA Gatekeeper Library • Open-Source Contributor • Rego / OPA • github.com/open-policy-agent/gatekeeper-library

- Contributed ConstraintTemplate enforcing mandatory backup labels on StatefulSet workloads, including Rego policy logic, unit tests, and documentation. Merged into the upstream Gatekeeper library for community use.

CERTIFICATIONS

Kubernetes & Cloud Native: Certified Kubernetes Administrator (CKA) • Certified Kubernetes Security Specialist (CKS) • Certified Kubernetes Application Developer (CKAD) • Kubernetes and Cloud Native Security Associate (KCSA) • Kubernetes and Cloud Native Associate (KCNA) • CNCF Kubestronaut

AWS & Azure: Azure Solutions Architect Expert (AZ-305) • Azure DevOps Engineer Expert (AZ-400) • Azure Administrator Associate (AZ-104) • AWS Solutions Architect Associate (SAA-C03)

Infrastructure & Other: HashiCorp Terraform Associate (004) • CCNA • VMCE • VMCA • VMware VCP-DCV

EDUCATION

University of Colorado Boulder • M.S. Computer Science

2024 – 2026

Graduate Certificates: Data Science, Artificial Intelligence

The Ohio State University • B.S.

2016